HTML MASTER NOTES

A Detailed Beginner Handbook

Concept • Syntax • Explanation • Working Code • Browser Output

A self-contained guide to learning HTML from its foundations through forms, semantic page structure, media, metadata and complete page examples. Detailed explanations, visual results and concise reference material.

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Each chapter explains the purpose of the concept, illustrates working HTML code and shows a screenshot rendered in a real browser. Some outputs, such as pickers and controls, vary by browser.

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01 HTML, CSS and JavaScript

Concept and purpose

HTML gives meaning and structure. CSS controls appearance. JavaScript adds behavior.

Detailed explanation

HTML is not a programming language; it is a markup language. A browser reads HTML and constructs a document.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<h1>My first website</h1>
<p>HTML creates the content.</p>
<p style="color: #087e8b">CSS changes its appearance.</p>
<button type="button">A button (no action yet)</button>
```

RENDERED BROWSER OUTPUT



Output explained: The browser shows a heading, two paragraphs and a button. Without JavaScript or form behavior, clicking the button does nothing.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

02 The complete HTML document

Concept and purpose

Start with a proper document skeleton in every .html file.

Detailed explanation

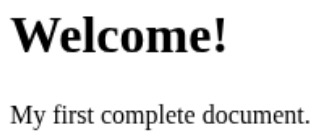
DOCTYPE triggers modern standards mode; html lang gives language; head holds metadata and body holds visible content.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<!doctype html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <title>My First Page</title>
</head>
<body>
  <h1>Welcome!</h1>
  <p>My first complete document.</p>
</body>
</html>
```

RENDERED BROWSER OUTPUT



Output explained: The tab title is My First Page; only the h1 and paragraph appear in the document area.

Important note: Always save files using the .html extension.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

03 Elements, tags and attributes

Concept and purpose

An element is made using tags, content and optional attributes.

Detailed explanation

Opening: ; closing: ; attribute examples: id, class, title, style, href, src, alt.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<h2 title="This is a tooltip">Hover over me</h2>
<p id="intro" class="description">A paragraph with attributes.</p>
<p style="color: seagreen">Inline CSS creates green text.</p>
```

RENDERED BROWSER OUTPUT



Output explained: Hovering the heading may show its title hint. The id and class are not visible by themselves.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

04 Headings and paragraphs

Concept and purpose

Use headings for document hierarchy rather than just font size.

Detailed explanation

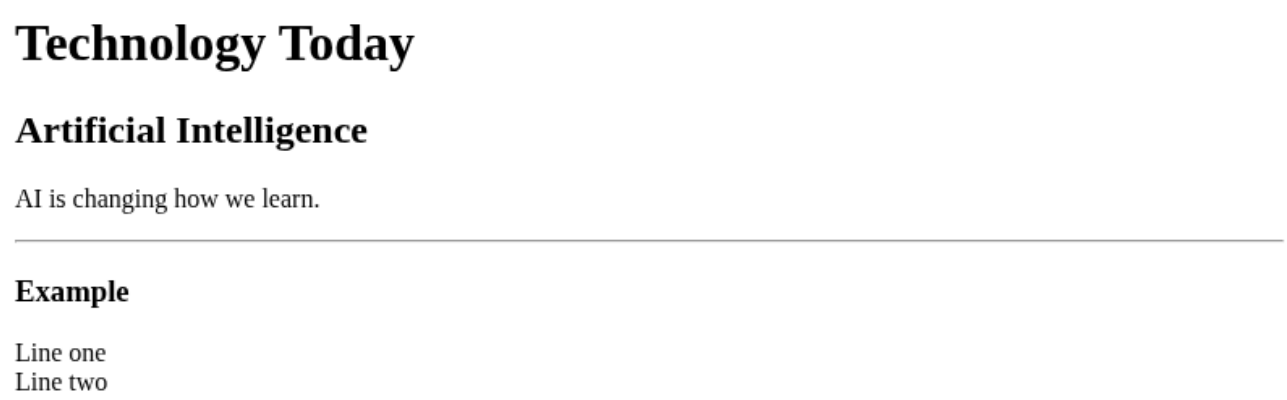
h1 is main topic; h2 subsection; h3 sub-subsection; p is a paragraph; br makes line break; hr makes thematic break.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<h1>Technology Today</h1>
<h2>Artificial Intelligence</h2>
<p>AI is changing how we learn.</p>
<hr>
<h3>Example</h3>
<p>Line one<br>Line two</p>
```

RENDERED BROWSER OUTPUT



Output explained: The heading levels differ in size by browser default. hr creates a divider; br starts another line.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

05 Text emphasis and formatting

Concept and purpose

Use formatting tags to give words clear meaning.

Detailed explanation

strong = importance; em = emphasis; mark = highlight; small = side note; del = deletion; ins = insertion; sup and sub = superscript/subscript.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<p><strong>Important:</strong> Submit today.</p>
<p>This is <em>emphasized</em> and <mark>highlighted</mark>.</p>
<p>Offer: <del>Rs 500</del> <ins>Rs 399</ins></p>
<p>H<sub>2</sub>O and x<sup>2</sup></p>
<small>Terms and conditions apply.</small>
```

RENDERED BROWSER OUTPUT

Important: Submit today.

This is *emphasized* and highlighted.

Offer: ~~Rs 500~~ Rs 399

H_2O and x^2

<small>Terms and conditions apply.</small>

Output explained: Important usually appears bold; emphasis italic; mark highlighted. Sup and sub shift text vertically.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

06 Quotes, abbreviations and code

Concept and purpose

Use meaningful elements for definitions, quotations and technical text.

Detailed explanation

abbr provides an expanded title; q is an inline quote; blockquote for longer quotations; code is computer code; kbd is a key.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<p><abbr title="HyperText Markup Language">HTML</abbr> is useful.</p>
<p>Teacher said <q>Practice every day</q>.</p>
<blockquote>Good structure improves readability.</blockquote>
<p>Write <code>&lt;p&gt;</code>. Press <kbd>Ctrl</kbd> + <kbd>S</kbd>.</p>
```

RENDERED BROWSER OUTPUT

HTML is useful.

Teacher said “Practice every day”.

> Good structure improves readability.

Write <p>. Press Ctrl + S.

Output explained: q displays quotation marks in most browsers; code and keyboard tags have a monospaced appearance.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

07 HTML entities and special characters

Concept and purpose

Some characters must be written using entities to be displayed literally.

Detailed explanation

< represents <; > represents >; & represents &; © represents copyright;   is a nonbreaking space.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<p>Write &amp;lt;h1&amp;gt; for a main heading.</p>
<p>Fish &amp;amp; Chips</p>
<p>&amp;copy; 2026 Learning Notes</p>
<p>Price: &amp;#8377;299</p>
```

RENDERED BROWSER OUTPUT

Write <h1> for a main heading.

Fish & Chips

© 2026 Learning Notes

Price: ₹299

Output explained: The page displays literal angle brackets, an ampersand, a copyright symbol and a rupee sign.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

08 Unordered, ordered and description lists

Concept and purpose

Lists make repeated information easier to read.

Detailed explanation

ul = bullets, ol = numbered sequence, li = list item, dl = definition list, dt = term, dd = description.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<h3>Skills</h3>
<ul><li>HTML</li><li>CSS</li><li>Python</li></ul>
<h3>Steps</h3>
<ol><li>Write code</li><li>Save file</li><li>Open browser</li></ol>
<dl><dt>HTML</dt><dd>Page structure</dd></dl>
```

RENDERED BROWSER OUTPUT

Skills

- HTML
- CSS
- Python

Steps

1. Write code
2. Save file
3. Open browser

HTML
: Page structure

Output explained: The browser shows bullet points, numbered steps and an indented definition.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

09 Links and external navigation

Concept and purpose

The anchor element creates hyperlinks to other resources.

Detailed explanation

href is link destination; target=_blank opens a new tab; rel=noopener is a safe external-link practice.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<p><a href="https://example.com">Visit a website</a></p>
<p><a href="https://example.com" target="_blank" rel="noopener">Open new tab</a></p>
<p><a href="mailto:hello@example.com">Send email</a></p>
<p><a href="tel:+911234567890">Call phone</a></p>
```

RENDERED BROWSER OUTPUT

Visit a website

Open new tab

Send email

Call phone

Output explained: Four underlined links appear. Their actions depend on network, browser and device support.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

10 Same-page navigation and IDs

Concept and purpose

Use an #id fragment to jump to a section in the same document.

Detailed explanation

A fragment link href=#contact matches an element with id=contact. IDs must be unique.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<nav>
  <a href="#about">About</a> |
  <a href="#contact">Contact</a>
</nav>
<section id="about"><h2>About</h2><p>Learning HTML.</p></section>
<section id="contact"><h2>Contact</h2><p>Email us.</p></section>
```

RENDERED BROWSER OUTPUT



Output explained: Clicking About or Contact navigates to the matching section (may barely scroll on a short page).

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

11 Linking two files and downloading

Concept and purpose

Links can point to files in the same folder, another folder or a download.

Detailed explanation

Relative paths are resolved from the current document. download suggests saving a same-origin file; it does not create that file.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<p><a href="about.html">Open the About page</a></p>
<p><a href="notes.txt" download>Download a text note</a></p>
<p><a href="../index.html">Go to parent folder index</a></p>
```

RENDERED BROWSER OUTPUT

Open the About page

Download a text note

Go to parent folder index

Output explained: The browser displays three links. In the companion files, about.html and notes.txt exist; parent link works only if that file exists.

Important note: Use ./image.jpg for same-folder assets and ../ for the parent directory.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

12 Images and alternative text

Concept and purpose

Show pictures using the img element; describe meaningful images with alt.

Detailed explanation

src specifies a URL/path. width and height reserve dimensions. alt is for accessibility or when images fail.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<h3>A simple illustration</h3>
<img src="sample.svg" alt="Blue mountain beneath a sun" width="280">
<p>Image is stored locally as sample.svg.</p>
```

RENDERED BROWSER OUTPUT

A simple illustration



Image is stored locally as sample.svg.

Output explained: The image displays a simple landscape illustration; its alt text provides an alternative description.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

13 Figures and captions

Concept and purpose

Use figure + figcaption when an illustration needs a caption.

Detailed explanation

figure represents self-contained content; figcaption provides its label.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<figure>
  <img src="sample.svg" alt="Blue mountains" width="260">
  <figcaption>Figure 1. A mountain illustration.</figcaption>
</figure>
```

RENDERED BROWSER OUTPUT



Figure 1. A mountain illustration.

Output explained: An illustration appears with a caption below it.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

14 Tables: rows, headings and cells

Concept and purpose

Use tables for facts arranged by rows and columns, not for whole-page layout.

Detailed explanation

table contains tr rows; th is a header cell; td is data cell; caption names a table.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<table border="1" cellpadding="7">
 <caption>Student Scores</caption>
 <tr><th>Subject</th><th>Marks</th></tr>
 <tr><td>HTML</td><td>92</td></tr>
 <tr><td>Python</td><td>85</td></tr>
</table>
```

RENDERED BROWSER OUTPUT

Student Scores

Subject	Marks
HTML	92
Python	85

Output explained: A two-column table displays subject names and marks.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

15 Tables: thead, tbody, tfoot

Concept and purpose

Group table header, body and summary semantically.

Detailed explanation

thead groups headings; tbody groups regular rows; tfoot groups totals.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<table border="1" cellpadding="7">
 <thead><tr><th>Item</th><th>Cost</th></tr></thead>
 <tbody>
  <tr><td>Domain</td><td>500</td></tr>
  <tr><td>Hosting</td><td>700</td></tr>
 </tbody>
 <tfoot><tr><th>Total</th><th>1200</th></tr></tfoot>
</table>
```

RENDERED BROWSER OUTPUT

Item	Cost
Domain	500
Hosting	700
Total	**1200**

Output explained: A header row, two items and a total row are shown.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

16 Tables: rowspan and colspan

Concept and purpose

Merge table cells horizontally or vertically when representing grouped data.

Detailed explanation

colspan=2 takes two columns; rowspan=2 spans two rows. The table grid must remain consistent.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<table border="1" cellpadding="8">
 <tr><th colspan="3">Semester Summary</th></tr>
 <tr><th>Term</th><th>Subject</th><th>Grade</th></tr>
 <tr><td rowspan="2">5</td><td>OS</td><td>A</td></tr>
 <tr><td>DBMS</td><td>B</td></tr>
</table>
```

RENDERED BROWSER OUTPUT

Semester Summary		
Term	**Subject**	**Grade**
5	OS	A
	DBMS	B

Output explained: The title spans all three columns and the term number spans two rows.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

17 Forms, labels and text fields

Concept and purpose

Forms collect user input; labels explain fields.

Detailed explanation

form wraps controls; label for matches input id; name identifies submitted field; required requests nonempty input.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<form>
 <label for="fullName">Full name:</label><br>
 <input id="fullName" name="fullName" type="text" placeholder="Enter your name"
required><br><br>
 <label for="email">Email:</label><br>
 <input id="email" name="email" type="email" required><br><br>
 <button type="button">Demo only</button>
</form>
```

RENDERED BROWSER OUTPUT



Output explained: Two text fields and a demo button appear. No backend submission is implemented.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

18 Password, date, time and number inputs

Concept and purpose

Input type changes the native browser control.

Detailed explanation

password masks characters; number offers numeric entry; date/time use browser pickers.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<form>
 <label>Password: <input type="password" minlength="6"></label><br><br>
 <label>Age: <input type="number" min="1" max="120"></label><br><br>
 <label>Date: <input type="date"></label><br><br>
 <label>Time: <input type="time"></label>
</form>
```

RENDERED BROWSER OUTPUT



Output explained: Password, number, date and time controls render using the browser’s native UI.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

19 Radio buttons and checkboxes

Concept and purpose

Radio = choose one from a named group; checkbox = independent on/off.

Detailed explanation

Radio controls with the same name form a choice group. Checkbox can mark multiple selections.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<fieldset>
 <legend>Preferred plan</legend>
 <label><input type="radio" name="plan" checked> Basic</label>
 <label><input type="radio" name="plan"> Premium</label>
</fieldset>
<p><label><input type="checkbox" checked> HTML</label>
<label><input type="checkbox"> CSS</label></p>
```

RENDERED BROWSER OUTPUT



Output explained: Only one plan can be selected. HTML and CSS checkboxes can be toggled independently.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

20 Dropdowns and multiple-line text

Concept and purpose

select provides options; textarea supports longer typed messages.

Detailed explanation

option is a choice within select; textarea rows/cols hint at visible size.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<label for="role">Role:</label>
<select id="role" name="role">
 <option value="frontend">Frontend</option>
 <option value="analyst">Data Analyst</option>
 <option value="ml">ML Engineer</option>
</select>
<p><label for="feedback">Feedback:</label></p>
<textarea id="feedback" rows="3" cols="36" placeholder="Write feedback here"></textarea>
```

RENDERED BROWSER OUTPUT

Role: Frontend

Feedback:

Write feedback here

Output explained: A dropdown and a larger editable textbox appear.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

21 Files, URLs and slider inputs

Concept and purpose

Special form controls allow file selection, websites and ranges.

Detailed explanation

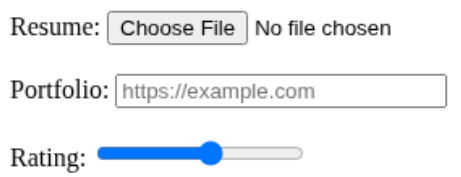
file only selects a local file; accept filters the file-picker choices. range represents a value within bounds.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<form>
 <label>Resume: <input type="file" accept=".pdf,.doc,.docx"></label><br><br>
 <label>Portfolio: <input type="url" placeholder="https://example.com"></label><br><br>
 <label>Rating: <input type="range" min="1" max="10" value="6"></label>
</form>
```

RENDERED BROWSER OUTPUT



Output explained: Browser displays a file picker, URL field and slider. Files are not uploaded without a processing endpoint.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

22 Validation, submission and buttons

Concept and purpose

HTML can perform basic input checks before submitting a form.

Detailed explanation

required, minlength, maxlength, min, max and pattern support client validation. type=submit sends; type=reset clears form.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<form onsubmit="event.preventDefault(); alert('Demo only: no server connected');">
 <label>Username: <input name="user" required minlength="3" maxlength="12"
></label><br><br>
 <label>Score: <input type="number" min="0" max="100" required></label><br><br>
 <button type="submit">Check & Submit</button>
 <button type="reset">Reset</button>
</form>
```

RENDERED BROWSER OUTPUT

Username:

Score:

Check & Submit Reset

Output explained: When submitted, browser validates entries. This self-contained demonstration stops real submission and shows a message.

Important note: HTML validation helps the user but does not replace server-side validation.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

23 Semantic page structure

Concept and purpose

Semantic tags describe what each region of the page means.

Detailed explanation

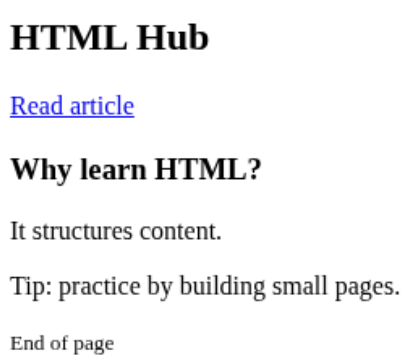
header, nav, main, section, article, aside and footer are meaningful containers.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<header><h2>HTML Hub</h2></header>
<nav><a href="#story">Read article</a></nav>
<main>
 <article id="story"><h3>Why learn HTML?</h3><p>It structures content.</p></article>
 <aside><p>Tip: practice by building small pages.</p></aside>
</main>
<footer><small>End of page</small></footer>
```

RENDERED BROWSER OUTPUT



Output explained: A heading, link, article, tip and footer appear in order.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

24 div, span, block and inline

Concept and purpose

Use general containers when there is no more meaningful semantic element.

Detailed explanation

div is usually block-level; span usually inline. CSS can change display behavior.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<div style="background:#e6f2ff;padding:10px">
 <h3>Notice</h3>
 <p>This is a <span style="color:crimson">special</span> message.</p>
</div>
<div>Another block starts below.</div>
```

RENDERED BROWSER OUTPUT

Notice

This is a special message.

Another block starts below.

Output explained: The div creates a colored region. span styles one word without starting a new line.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

25 details and summary: built-in disclosure

Concept and purpose

Create a show/hide section with HTML alone.

Detailed explanation

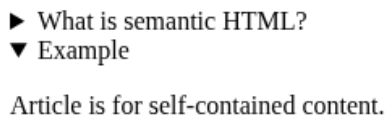
details toggles open/closed; summary is the clickable label.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<details>
 <summary>What is semantic HTML?</summary>
 <p>HTML tags that express the meaning of content.</p>
</details>
<details open>
 <summary>Example</summary>
 <p>Article is for self-contained content.</p>
</details>
```

RENDERED BROWSER OUTPUT



Output explained: First disclosure starts closed; second begins expanded. Click each summary to toggle.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

26 Metadata: title, description and viewport

Concept and purpose

Metadata belongs in head and helps browsers and search engines interpret the page.

Detailed explanation

title names tab; description can be used in search snippets; viewport aids mobile presentation; robots gives crawl directives.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<!doctype html>
<html lang="en"><head>
 <meta charset="UTF-8">
 <meta name="viewport" content="width=device-width,initial-scale=1">
 <meta name="description" content="Beginner HTML practice notes">
 <meta name="robots" content="index,follow">
 <meta name="theme-color" content="#123456">
 <title>Metadata Demo</title>
</head><body><h2>Metadata demo</h2>
<p>Metadata usually is not visible in the page body.</p>
</body></html>
```

RENDERED BROWSER OUTPUT



Output explained: Tab title changes to Metadata Demo; meta tags do not add visible body content.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

27 Canonical URLs, Open Graph and link tags

Concept and purpose

A published site can describe its preferred URL and sharing preview.

Detailed explanation

canonical link identifies preferred URL; og:title, og:description and og:image inform many social previews; preconnect is a performance hint.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<head>
 <title>Example Article</title>
 <link rel="canonical" href="https://example.com/article">
 <meta property="og:title" content="Example Article">
 <meta property="og:description" content="An HTML tutorial">
 <meta property="og:type" content="article">
 <meta property="og:url" content="https://example.com/article">
 <link rel="preconnect" href="https://fonts.googleapis.com">
</head>
<body><h2>Sharing metadata</h2><p>See page source for the tags.</p></body>
```

RENDERED BROWSER OUTPUT



Output explained: Heading and paragraph appear; sharing metadata is not directly visible inside the page. Replace example URLs when publishing.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

28 Audio, video and iframe basics

Concept and purpose

HTML can embed media and external content.

Detailed explanation

audio and video can provide native controls; iframe embeds another page, subject to the external site allowing it.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

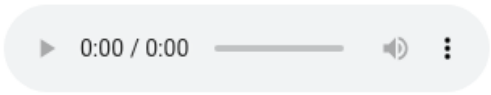
WORKING HTML CODE

```
<h3>Media elements</h3>
<p>Audio (requires a real audio file):</p>
<audio controls><source src="sample-audio.mp3" type="audio/mpeg"></audio>
<p>Video (requires a real video file):</p>
<video controls width="260"><source src="sample-video.mp4" type="video/mp4"></video>
```

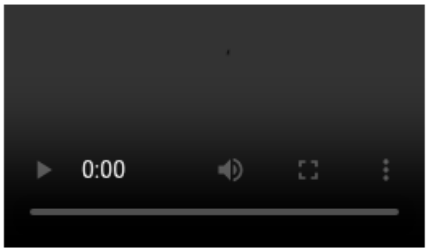
RENDERED BROWSER OUTPUT

Media elements

Audio (requires a real audio file):



Video (requires a real video file):



Output explained: Native audio and video controls appear; playback is unavailable until matching media files are added.

Important note: Only use media you own or have permission to publish.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

29 Accessible HTML and common mistakes

Concept and purpose

Correct markup makes projects easier to navigate and maintain.

Detailed explanation

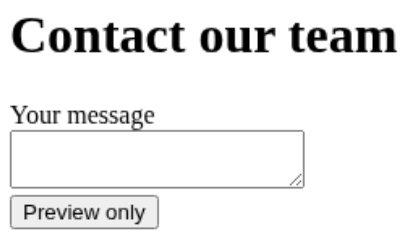
Use label + id, descriptive alt, unique IDs, real headings, relevant button type and keyboard-friendly links.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<main>
 <h1>Contact our team</h1>
 <form>
  <label for="message">Your message</label><br>
  <textarea id="message" name="message" rows="2" required></textarea><br>
  <button type="button">Preview only</button>
 </form>
</main>
```

RENDERED BROWSER OUTPUT



Output explained: An accessible label, textarea and button are shown. Real submission is not configured.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

30 Complete example: movie landing page

Concept and purpose

Combine a header, navigation, movie cards and a small pricing table.

Detailed explanation

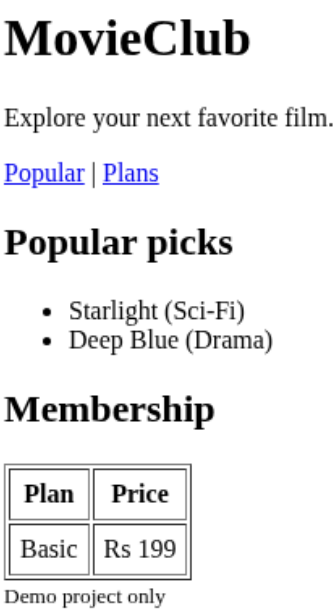
A complete standalone example combining meaningful layout, links and a data table.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<header><h1>MovieClub</h1><p>Explore your next favorite film.</p></header>
<nav><a href="#popular">Popular</a> | <a href="#plans">Plans</a></nav>
<main>
 <section id="popular"><h2>Popular picks</h2>
  <ul><li>Starlight (Sci-Fi)</li><li>Deep Blue (Drama)</li></ul>
 </section>
 <section id="plans"><h2>Membership</h2>
  <table border="1" cellpadding="6"><tr><th>Plan</th><th>Price</th></tr>
  <tr><td>Basic</td><td>Rs 199</td></tr></table>
 </section>
</main><footer><small>Demo project only</small></footer>
```

RENDERED BROWSER OUTPUT



Output explained: A movie-site-style home page appears with navigation, content and a price table.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

31 Complete example: result statement

Concept and purpose

Practice tables and summary information with a university-style document.

Detailed explanation

A complete sample document using fictional names and sample marks.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<header><h1>Campus Results</h1><p>Sample score statement</p></header>
<section><h2>Candidate</h2><p>Name: Student A | Semester: 2</p></section>
<table border="1" cellpadding="7">
 <tr><th>Course</th><th>Score</th><th>Status</th></tr>
 <tr><td>HTML</td><td>86</td><td>Pass</td></tr>
 <tr><td>Networks</td><td>81</td><td>Pass</td></tr>
</table>
<p><strong>Total:</strong> 167 / 200</p>
```

RENDERED BROWSER OUTPUT

Campus Results

Sample score statement

Candidate

Name: Student A | Semester: 2

Course	Score	Status
HTML	86	Pass
Networks	81	Pass

Total: 167 / 200

Output explained: A formal result document contains candidate information, a score table and a total.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

32 Complete example: application portal

Concept and purpose

Combine multiple form input types in a structured single page.

Detailed explanation

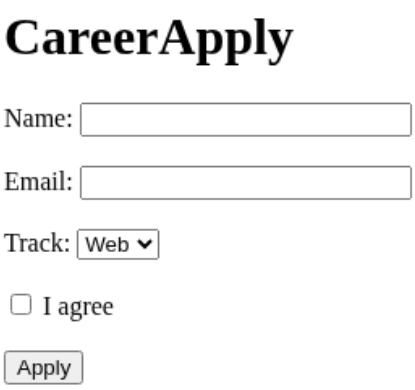
A complete standalone example that illustrates several HTML controls.

The markup in the example below is interpreted by the browser as document structure. HTML describes the meaning and relationships of content; visual presentation is governed by browser defaults and any CSS rules. Understanding the role of each element is more important than memorising its appearance.

WORKING HTML CODE

```
<h1>CareerApply</h1>
<form onsubmit="event.preventDefault(); alert('Practice form: no backend');">
 <label for="name">Name:</label> <input id="name" required><br><br>
 <label for="mail">Email:</label> <input id="mail" type="email" required><br><br>
 <label for="track">Track:</label>
 <select id="track"><option>Web</option><option>Data</option></select><br><br>
 <label><input type="checkbox" required> I agree</label><br><br>
 <button type="submit">Apply</button>
</form>
```

RENDERED BROWSER OUTPUT



Output explained: Shows name, email, track selection, required agreement and apply button. Submission is a local demo only.

Remember: HTML elements provide structure and meaning. Interactive behavior such as saving form data, account login or uploading files requires an appropriate processing service; appearance can be customized separately with CSS.

HTML Reference and Common Mistakes

Document structure

Use a correct DOCTYPE, html, head and body. The document title belongs inside head, while visible headings belong inside body.

Proper nesting

Close nested tags in the reverse order in which they were opened. Some elements such as img, input, meta, link, br and hr are void elements.

Use meaningful HTML

Use h1 through h6 for heading hierarchy and semantic elements such as article, section and nav for their intended meanings.

Links and assets

Relative paths depend on the folder containing the HTML file. Broken paths prevent images and linked documents from loading.

Tables

Use th for header cells, td for data, caption to describe the table, and thead/tbody/tfoot for row groups.

Forms

Associate each input with a label. Give submission controls a name, choose the correct type, and remember that required is only browser-side validation.

Images and accessibility

Use descriptive alternative text for meaningful images. Keep link text clear and ensure forms have accessible labels.

Metadata

Set charset, viewport, title and a meaningful description. Meta descriptions and robots directives do not guarantee search-engine rankings.

CSS and JavaScript

HTML is the structure, CSS styles the content, and JavaScript is responsible for custom behavior. Inline styles work but a stylesheet is more maintainable.

Publishing

Static HTML pages work locally. Publishing online requires hosting. Features involving authentication, storage, uploads or backend validation need additional services.

Closing Summary

HTML is a semantic markup language for organising information on the web. Start with an accessible document skeleton, choose elements for their meaning, use tables only for tabular data and forms for input, and verify how every page is rendered by a browser. Correct structure is the foundation of effective web development.